

Ragionato per 12s

TSIA2 Math Practice Test 4

PART A — QUESTIONS

TSIA2-M4-001

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A bookstore sells a used calculator for \$38.40. This price is 80% of the original price. What was the original price of the calculator?

A) \$30.72

B) \$46.40

C) \$48.00

D) \$50.40

TSIA2-M4-002

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which value of x satisfies the equation $4x + 7 = 31 - 2x$?

A) 3

B) 4

C) 5

D) 6

TSIA2-M4-003

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular shipping label is 8 centimeters long and 5 centimeters wide. A smaller similar label has a width of 3 centimeters. What is the length of the smaller label?

A) 4.2 centimeters

B) 4.8 centimeters

C) 5.0 centimeters

D) 6.0 centimeters

TSIA2-M4-004

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The table shows the number of hours four employees worked last week.

Employee A: 18

Employee B: 24

Employee C: 27

Employee D: 31

What is the mean number of hours worked?

- A) 24
- B) 25
- C) 26
- D) 27

TSIA2-M4-005

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A car travels 238 miles using 8.5 gallons of gasoline. What is the car's fuel efficiency in miles per gallon?

- A) 24 miles per gallon
- B) 26 miles per gallon
- C) 28 miles per gallon
- D) 30 miles per gallon

TSIA2-M4-006

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A value of y is 5 more than twice a value of x . Which equation represents this relationship?

- A) $y = 2x - 5$
- B) $y = 2x + 5$
- C) $y = 5x + 2$
- D) $y = x + 10$

TSIA2-M4-007

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A rectangular prism has a length of 9 inches, a width of 4 inches, and a height of 6 inches. What is its volume?

- A) 19 cubic inches
- B) 108 cubic inches
- C) 216 cubic inches
- D) 324 cubic inches

TSIA2-M4-008

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A drawer contains 7 black socks and 5 white socks. One sock is selected at random. What is the probability that the sock is white?

- A) $\frac{5}{12}$
- B) $\frac{7}{12}$
- C) $\frac{5}{7}$
- D) $\frac{1}{2}$

TSIA2-M4-009

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A student completed 45 questions on a review assignment. This was $\frac{3}{5}$ of the total number of questions on the assignment. How many total questions were on the assignment?

- A) 60
- B) 72
- C) 75
- D) 90

TSIA2-M4-010

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

A sequence begins with 40 and decreases by 6 each term. Which expression gives the n th term of the sequence?

- A) $40 - 6n$
- B) $46 - 6n$
- C) $40 + 6n$
- D) $6n - 40$

TSIA2-M4-011

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A circle has an area of 49π square meters. What is the radius of the circle?

- A) 7 meters
- B) 14 meters
- C) 24.5 meters
- D) 49 meters

TSIA2-M4-012

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

The waiting times, in minutes, for six customers were 12, 15, 15, 18, 20, and 28. Which statement is true?

- A) The median is 15 minutes.
- B) The median is 16.5 minutes.
- C) The range is 15 minutes.
- D) The mode is 18 minutes.

TSIA2-M4-013

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A campus printer charges \$0.09 per black-and-white page and \$0.35 per color page. A student prints 24 black-and-white pages and 6 color pages. What is the total cost?

A) \$3.06

B) \$3.96

C) \$4.26

D) \$4.86

TSIA2-M4-014

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Solve the inequality $3x - 4 > 11$.

A) $x > 5$

B) $x > 7/3$

C) $x < 5$

D) $x < 7/3$

TSIA2-M4-015

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A right triangle has a hypotenuse of 25 inches and one leg of 7 inches. What is the length of the other leg?

A) 18 inches

B) 20 inches

C) 24 inches

D) 32 inches

TSIA2-M4-016

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A quality check of 200 light bulbs found that 12 bulbs were defective. Based on this sample, what percent of the bulbs were defective?

A) 4%

B) 6%

C) 8%

D) 12%

TSIA2-M4-017

Section: Math

Type: mcq

Category: Quantitative Reasoning

Prompt:

A water tank contains 18 gallons of water. Water is added at a constant rate for 4 minutes, increasing the amount in the tank to 31 gallons. What is the average rate at which water is added?

- A) 2.75 gallons per minute
- B) 3.25 gallons per minute
- C) 4.50 gallons per minute
- D) 7.75 gallons per minute

TSIA2-M4-018

Section: Math

Type: mcq

Category: Algebraic Reasoning

Prompt:

Which expression is equivalent to $(x - 3)(x + 8)$?

- A) $x^2 + 5x - 24$
- B) $x^2 - 5x - 24$
- C) $x^2 + 11x - 24$
- D) $x^2 + 5x + 24$

TSIA2-M4-019

Section: Math

Type: mcq

Category: Geometric and Spatial Reasoning

Prompt:

A line segment has endpoints $(-3, -2)$ and $(5, 4)$. What is the midpoint of the segment?

- A) $(1, 1)$
- B) $(2, 2)$
- C) $(-1, -3)$
- D) $(4, 6)$

TSIA2-M4-020

Section: Math

Type: mcq

Category: Probabilistic and Statistical Reasoning

Prompt:

A password system uses one letter followed by one digit. The letter can be A, B, C, or D, and the digit can be 1, 2, or 3. If each password is equally likely, what is the probability of randomly choosing the password C2?

- A) $1/7$
- B) $1/12$
- C) $2/12$
- D) $1/24$

PART B — ANSWER KEY + EXPLANATIONS

TSIA2-M4-001 — Correct: C — Correct Answer: \$48.00 — Explanation: If \$38.40 is 80% of the original price, then the original price is $38.40 \div 0.80 = \$48.00$.

TSIA2-M4-002 — Correct: B — Correct Answer: 4 — Explanation: Add $2x$ to both sides to get $6x + 7 = 31$. Subtract 7 to get $6x = 24$, so $x = 4$.

TSIA2-M4-003 — Correct: B — Correct Answer: 4.8 centimeters — Explanation: The scale factor from the larger label to the smaller label is $3/5$. The smaller length is $8 \times 3/5 = 4.8$ centimeters.

TSIA2-M4-004 — Correct: B — Correct Answer: 25 — Explanation: The total number of hours is $18 + 24 + 27 + 31 = 100$. Dividing by 4 gives a mean of 25.

TSIA2-M4-005 — Correct: C — Correct Answer: 28 miles per gallon — Explanation: Divide miles by gallons: $238 \div 8.5 = 28$ miles per gallon.

TSIA2-M4-006 — Correct: B — Correct Answer: $y = 2x + 5$ — Explanation: “Twice a value of x ” is $2x$, and “5 more than” that amount is $2x + 5$.

TSIA2-M4-007 — Correct: C — Correct Answer: 216 cubic inches — Explanation: The volume of a rectangular prism is length $\times$ width $\times$ height, so $9 \times 4 \times 6 = 216$ cubic inches.

TSIA2-M4-008 — Correct: A — Correct Answer: $5/12$ — Explanation: There are $7 + 5 = 12$ socks total, and 5 are white. The probability is $5/12$.

TSIA2-M4-009 — Correct: C — Correct Answer: 75 — Explanation: If 45 is $3/5$ of the total, then the total is $45 \div 3/5 = 45 \times 5/3 = 75$.

TSIA2-M4-010 — Correct: B — Correct Answer: $46 - 6n$ — Explanation: For an arithmetic sequence with first term 40 and common difference -6 , the n th term is $40 + (n - 1)(-6)$, which simplifies to $46 - 6n$.

TSIA2-M4-011 — Correct: A — Correct Answer: 7 meters — Explanation: The area of a circle is $A = \pi r^2$. If $\pi r^2 = 49\pi$, then $r^2 = 49$, so $r = 7$ meters.

TSIA2-M4-012 — Correct: B — Correct Answer: The median is 16.5 minutes. — Explanation: With six values, the median is the average of the third and fourth values: $(15 + 18) \div 2 = 16.5$.

TSIA2-M4-013 — Correct: C — Correct Answer: \$4.26 — Explanation: The black-and-white pages cost $24 \times \$0.09 = \2.16 , and the color pages cost $6 \times \$0.35 = \2.10 . The total is \$4.26.

TSIA2-M4-014 — Correct: A — Correct Answer: $x > 5$ — Explanation: Add 4 to both sides to get $3x > 15$. Divide by 3 to get $x > 5$.

TSIA2-M4-015 — Correct: C — Correct Answer: 24 inches — Explanation: Using the Pythagorean theorem, the missing leg is $\sqrt{(25^2 - 7^2)} = \sqrt{(625 - 49)} = \sqrt{576} = 24$ inches.

TSIA2-M4-016 — Correct: B — Correct Answer: 6% — Explanation: The defective rate is $12/200 = 0.06$, or 6%.

TSIA2-M4-017 — Correct: B — Correct Answer: 3.25 gallons per minute — Explanation: The amount of water increased by $31 - 18 = 13$ gallons. Over 4 minutes, the rate was $13 \div 4 = 3.25$ gallons per minute.

TSIA2-M4-018 — Correct: A — Correct Answer: $x^2 + 5x - 24$ — Explanation: Multiply the binomials: $x^2 + 8x - 3x - 24 = x^2 + 5x - 24$.

TSIA2-M4-019 — Correct: A — Correct Answer: (1, 1) — Explanation: The midpoint is $((-3 + 5)/2, (-2 + 4)/2) = (1, 1)$.

TSIA2-M4-020 — Correct: B — Correct Answer: $1/12$ — Explanation: There are $4 \times 3 = 12$ possible passwords, and only one is C2. The probability is $1/12$.